

Experimental And Theoretical Research On Corrosion Inhibitive Behavior Of A Synthesized Schiff Base On Copper Corrosion In Acid Media.

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Abstract

The electrochemical analysis and the theoretical research suggested that a Schiff base, 1,3-bis((pyridin-2-ylmethylene)thiourea, which is obtained by the reaction of 2-formyl pyridine and thiourea serves as one of the effective inhibitors for the corrosion of copper in the 1.0 M HNO₃ and 0.5 M H₂SO₄ acid solutions. It was observed that the inhibitory effect of 1,3-bis((pyridin-2-ylmethylene)thiourea on both corrosive environments was of mixed nature with slight excess cathodic based. The relationship that exists between the temperature, and the rate of corrosion was investigated in the temperature range 35 to 55° C and the activation energy (E_a) of the reaction was obtained. The quantity of the molecules of the Schiff base adsorbed on the copper surface was as predicted by Langmuir adsorption isotherm. The studies on the potential of the zero charge (PZC) have been completed in order to comprehend the mechanism of the adsorption of the Schiff base molecules on the metal surface. We have obtained the thermodynamic variables including adsorption equilibrium constant (K_{ads}) and the free energy of the adsorption (ΔG_{ads}). This value of adsorption free energy lies between -20kJmol⁻¹ to 40kJmol⁻¹ and the closest value of -40kJmol⁻¹ indicates that the adsorption mechanism is by the electrostatic coulombic attraction and chemisorption. The addition of the inhibitor to the corrosive media, the value of E_a decreased which indicates that the inhibitor chemisorbed onto the metal surface.

Keywords: Corrosion of copper; Schiff Base; Electrochemical Impedance Spectroscopy; Density Functional Theory; Potential of Zero Charge.

1.Introduction

Corrosion of copper in acidic media is an electrochemical process that degrades metal performance [1]. Sulfuric and nitric acids are common industrial corrosive media that aggressively attack copper surface [2]. H₂SO₄ promotes anodic dissolution and sulfate complex formation with copper ions [3]. HNO₃ is both an oxidizing and acidic environment, often accelerating copper passivation or pitting depending on concentration [4]. Controlling copper corrosion in these media is essential for electrical, plumbing, and chemical process reliability. Organic inhibitors are widely used because they are effective at low concentrations and form protective films [5]. Organic compounds with an azomethine (C=N-) bond (also known as Schiff bases) are potential corrosion inhibitors [6]. The inhibitor ability stems out of having heteroatoms (N, O, S) and π electrons structures that adsorb onto

metal surfaces. [7]. Adsorption produces barriers to active corrosion sites and limits the metal dissolution and cathodic reactions [8]. Schiff bases can coordinate with Cu⁺/Cu²⁺, forming complexes that stabilize the protective layer [9]. Molecular planarity, aromaticity, and substituent electron-donating/withdrawing groups affect adsorption strength [10]. Formation of Schiff base is generally found through condensing a primary amine and an aldehyde or a ketone [11]. The synthesized compound can be tailored to enhance solubility and surface affinity for copper [12]. Inhibitors assessment is typically performed with assistance of weight-loss, potentiodynamic polarization and electrochemical impedance spectroscopy (EIS). Thermodynamic parameters (ΔG°_{ads} , ΔH°_{ads}) can be used to differentiate between physisorption and chemisorption processes [13]. Most of the Schiff base molecules are of mixed type, inhibiting

corrosion reactions both at cathodic and anodic regions [14]. In H_2SO_4 , adsorption may compete with aggressive sulfate anions for surface sites; tailored Schiff bases can outcompete sulfate [15]. In HNO_3 , oxidizing conditions may modify inhibitor molecules or favor formation of passivating layers; inhibitor design must consider oxidative stability [16]. Optimal inhibition often occurs at a critical concentration where surface coverage approaches unity [17]. Quantum chemical calculations (DFT) correlate frontier orbital energies and charge distribution with inhibition efficiency [18]. Lower HOMO–LUMO gap and higher dipole moment generally indicate stronger adsorption propensity [19]. Langmuir, Temkin, and Frumkin isotherms are used to model adsorption behaviour on copper. A robust inhibitor should be eco-friendly, inexpensive, and synthetically accessible [20]. Recent studies report high inhibition efficiencies (>80–90%) for certain Schiff bases in acidic media [21]. Combining experimental electrochemistry with surface analysis provides mechanistic insight and validates inhibitor performance [22]. For copper, the inhibitor film must be stable under the specific potential regime of the metal in acid [23]. Metal–inhibitor complex formation can be confirmed by spectroscopic shifts (UV-Vis, FTIR) and XPS binding energies. Temperature studies reveal whether adsorption is chemisorption or exothermic physisorption [24]. The impedance of Electrochemical is frequently encountered to be having a greater charge transfer resistance and greater double-layer capacitance following inhibition. The polarization curves often suggest that Schiff bases are mixed inhibitors that have either weak anodic or cathodic characteristics. Inhibitor performance may decline under strongly oxidizing nitric acid unless stabilized by electron-withdrawing substituents [25]. Designing Schiff bases with chelating groups improves film cohesion and resistance to desorption [26]. Heteroatom-rich backbones (thiazole, pyridine, imidazole) often enhance adsorption and complexation with copper [27]. Green synthesis routes and biodegradable precursors are an active trend to reduce environmental impact [28]. Synergistic effects with halide ions or metal cations can further improve inhibition in some systems [29]. Reproducible lab protocols include standardized specimen preparation, solution aeration control, and temperature regulation. Comparison with benchmark inhibitors (e.g., benzotriazole for copper) contextualizes Schiff base performance. Long-term immersion tests assess durability and film repair ability under continuous exposure. Corrosion rate reduction is quantified as percent inhibition efficiency from weight-loss or electrochemical data. Mechanistic models propose initial physisorption followed by chemisorption and complex formation at

higher coverage. Surface morphology after inhibition often shows smoother, less-pitted copper compared with uninhibited samples. For practical deployment, scale-up, cost, and compatibility with process chemistry must be evaluated. Regulatory and environmental assessments are necessary before industrial adoption of new inhibitors. Overall, synthesized Schiff bases are versatile inhibitors that offer tunable chemistry for copper in H_2SO_4 and HNO_3 . Combining synthesis, electrochemical testing, surface analyses, and theory yields a complete evaluation framework. Future work should prioritize oxidative stability in nitric acid, greener synthesis, and real-world trials. This approach supports safer, cost-effective corrosion management for copper components in acidic environments. The synthesized Schiff base therefore represents a rational strategy for inhibiting copper corrosion in both H_2SO_4 and HNO_3 . Rigorous experimental validation complemented by computational insight will confirm its practical utility.

2. Experiments

2.1 Materials

For weight-loss measurements, copper specimens measuring 2.5 cm X 1 cm X 0.1 cm and composed of 0.002 weight percent Ni, 0.020 weight percent Al, 0.001 weight percent Mn, and 0.116 weight percent Si and rest being Cu were utilized. These specimens were degreased with acetone, dried, and polished in succession using emery sheets of grades 1/0, 2/0, 3/0, 4/0, and 5/0. To study the polarization and electrochemical impedance, a Teflon coated cylindrical copper electrode such that the exposed area is 0.2826 cm^2 and has the composition discussed above was used. Prior to these measurements the surface was cleaned with acetone and dried using nitrogen and after polished to mirror finish using the succession of the emery sheets of the various grades in increasing order (1/0, 2/0, 3/0, 4/0, and 5/0). After this treatment, water was considered to have a positive effect of surface cleanliness when it became perfectly uniformly wettable. Every experiment was conducted at $301 \pm 1\text{K}$, room temperature. All the solutions were prepared with the help of double distilled water and all the chemicals are of an analytical reagent grade (Sigma Aldrich), E Merck (India), to prepare stock solutions of the different acids.

2.2 Synthesis of Schiff base

Ethanol was added in minimum amount to dissolve 2-formylpyridine and thiourea in 2: 1 mole ratio. The 2 solutions were mixed and stirred for 6 h at room temperature. The solution was then precipitated and allowed to dry at room temperature. The precipitated Schiff base 1,3-bis ((pyridin-2-ylmethylene)thiourea washed using ethanol and dried. Figure.1

demonstrates the scheme and structure of synthesized Schiff base.

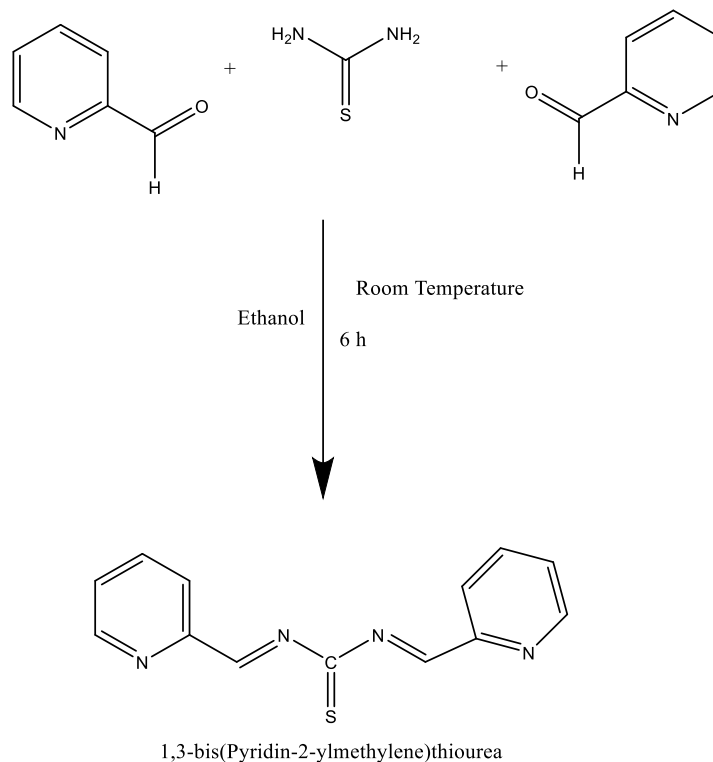


Figure.1 Synthesis of 1,3-bis(pyridin-2-ylmethylene)thiourea

The reaction of an aldehyde and thiourea forms a Schiff base is confirmed by the presence of vibrations representing C=N at 1658 cm⁻¹. Observation of

bands at 1546 cm⁻¹ and 1506 cm⁻¹ is related to the presence of the vibrations of C=C and C=N at the aromatic rings.

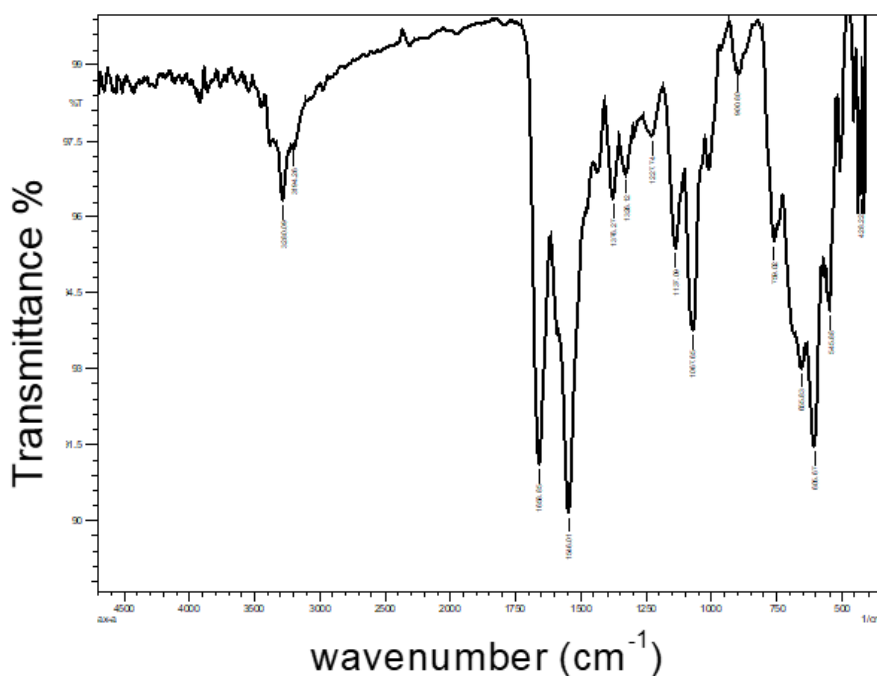


Figure.2 FTIR spectrum of Synthesized Schiff Base

2.3 Weightloss measurements

Weight loss was measured by following the American Society for Testing and Materials (ASTM) standard protocol. The copper specimens were completely

immersed for 2 hours in the 100 mL of acid solutions with and without the inhibitor at room temperature (301 ± 1 K). The inhibitory efficiency was

determined using the following formula based on the average weight loss of the two specimens:

$$IE = \left(\frac{W - W'}{W} \right) \times 100 \quad (i)$$

W and W' is the weightloss in uninhibited and inhibited solutions respectively.

2.4 Electrochemical Impedance Measurement

During the electrochemical studies, three electrodes set up was employed with the saturated calomel electrode(SCE) being used as reference electrode and Pt foil used as auxiliary electrode. The working electrode was a Teflon coated copper rod. A 45-minute stabilization period was allowed after the specimen had been immersed in the corrosive media before the impedance was measured. It was demonstrated that this was sufficient to stabilize E_{ocp} Vs (SCE). The electrochemical impedance measurements have been conducted at open circuit potential (E_{ocp} Vs (SCE)). The electrochemical impedance measurements were conducted using computer controlled potentiostat (Zahner Zennium XC high-grade electrochemical Instrument). The analysis of the results was done using ZMAN impedance analysis software. Peak to peak sine excitation wave of 10mV was used and the frequency of AC scanned at 10mHz to 100KHz. Inhibition efficiencies (IE%) were calculated using the following formula [30]:

$$IE\% = \frac{R'_{ct} - R_{ct}}{R_{ct}} \times 100 \quad (ii)$$

Where R_{ct} and R'_{ct} are the values of charge transfer resistances in the inhibited and bare solutions respectively.

2.5 Potentiodynamic Polarization measurement

The identical cell configuration was used to record potentiodynamic polarization curves at a potential sweep rate of 1.6 mVs⁻¹. Through corrosion potential, the potentials were scanned mostly from more negative potential than OCP to more positive potential than OCP. The efficiencies of inhibition were determined assisted by the relationship [30-33].

$$IE\% = \frac{i_{corr} - i'_{corr}}{i_{corr}} \times 100 \quad (iii)$$

where the density of the corrosion current of the sample with an inhibitor and without an inhibitor are represented by the following variables i_{corr} and i'_{corr} .

2.6 Potential of Zero Charge

Electrochemical impedance spectra were recorded at different applied DC potential at 20 kHz frequency AC. In order to determine the potential of zero charge, values of the double layer capacitance against DC potentials were plotted.

et al 2006).

2.7 Effect of Temperature

To ascertain the activation energy and comprehend the process of inhibitor adsorption onto the copper surface, potentiodynamic polarization experiments were conducted in both acid solutions at temperatures between 35 and 55°C with and without the inhibitor. The Arrhenius equation is used to determine how corrosion rate varies with temperature [30,34] .

$$i_{corr} = A e^{-E_a/RT} \quad (iv)$$

where i_{corr} = density of the current density of corrosion, A= the Arrhenius pre-exponential constant, R= universal gas constant, E_a =energy of activation, T= absolute temperature.

2.8 Synergistic Effect

The corrosive inhibitive performance of the inhibitor and halide ions together was also investigated. The Aramki and Hackermann equation was used to assess the synergism parameter [34–37].

$$S_I = \frac{1 - I_{1+2}}{1 - I'_{1+2}} \quad (v)$$

Where $I_{1+2} = (I_1 + I_2)$; I_1 is the efficiency of the inhibition exerted by the halide, I_2 is the efficiency of the inhibition exerted by the inhibitor and I'_{1+2} is the efficiency of the inhibition exerted by the inhibitor affinity with the halide ion.

2.9 Computational Quantum Studies

Geometry optimization is performed in the gas phase and without any form of symmetry constraints with the 6-31G(d,p) base set [38,39] and hybrid functional B3LYP [40,41] using the Gaussian 98 algorithm [33]. After deciding the geometry in the optimization stage, the vibrational analysis is performed, and the number of imaginary frequencies is zero, which means that the resulting geometry is a real minimum on the potential energy surface [42].

3. Results and Discussions

3.1 Weightloss measurement

Table 1 displays the mild steel inhibition effectiveness values in 1.0 M HNO₃ and 0.5 M H₂SO₄ acid media at various inhibitor concentrations. It is evident that when the inhibitor concentration increased in both the acid solutions, the inhibition efficiency also increased. This behavior can be explained based on the strong interaction of the inhibitor molecules with the metal surface resulting in adsorption. As the concentration of inhibitor increases, the degree of adsorption also increases which is indicated by the increase in the effectiveness of inhibition. By creating a protective coating that is adsorbed to the metal surface and keeps it apart from the corrosive media, organic inhibitors often prevent metal dissolution [33].

Table.1 Corrosion inhibition efficiency of 1, 3-burydin- 2-ylmethylene thiourea on corrosion of copper in higher concentration of 1.2 M HNO₃ and 0.5 M acid of H₂SO₄.

Medium	Concentration of the Inhibitor (mM)	Corrosion Rate (mmpy)	Inhibition Efficiency (IE%)
1.0 M HNO ₃	Blank	48.02	-
	0.02	30.12	37.3
	0.05	19.24	59.9
	0.1	13.05	72.8
	0.5	9.112	81.0
	1.0	2.614	94.6
0.5 M H ₂ SO ₄	Blank	23.14	-
	0.02	9.556	58.7
	0.05	8.793	62.0
	0.1	5.739	75.2
	0.5	2.569	88.9
	1.0	1.134	95.1

3.2 Electrochemical Impedance Spectroscopy

Figures 3a and 3b display the impedance spectra at 290 K in 1.0 M nitric acid and 0.5M sulfuric acid solutions respectively, with and without inhibitor at

different concentrations. Table.2 represents the respective parameters of the electrochemical impedance.

Table.2 Electrochemical Impedance parameters of copper corrosion in 1.0 M HNO₃ and 0.5 M H₂SO₄ acid solutions in the absence of presence of Schiff base at various concentrations.

Medium	Inhibitor Concentration (mM)	R _s	R _{ct} Ωcm ⁻²	Y ₀ (X 10 ⁻⁶) Ωcm ⁻²	n	C _{dl} μFcm ⁻²	W	Inhibitor Efficiency (IE%)
1.0M HNO ₃	BLANK	1.124	91.1	871.0	0.920	157	24X10 ⁻³	-
	0.02	1.114	154.4	144.5	0.944	116	33X10 ⁻³	41
	0.05	1.227	293.8	96.72	0.954	94	67X10 ⁻³	69
	0.10	1.229	350.4	82.55	0.982	79	81X10 ⁻³	74
	0.50	1.248	828.2	78.67	0.971	61	92X10 ⁻³	89
	1.0	1.251	1518	42.36	0.919	44	103X10 ⁻³	94
0.5M H ₂ SO ₄	BLANK	1.357	102.7	426.1	0.953	224	35X10 ⁻³	-
	0.02	1.336	250.5	115.8	0.941	162	49X10 ⁻³	59
	0.05	1.347	285.3	96.22	0.989	108	68X10 ⁻³	64
	0.10	1.355	446.5	87.14	0.987	93	94X10 ⁻³	77
	0.50	1.421	855.8	65.28	0.950	75	109X10 ⁻³	88
	1.0	1.339	2054	46.23	0.961	36	118X10 ⁻³	95

The fact that depressed semicircles are developed is postulated to be due to non-ideal capacitive functionality of the electrochemical solid / liquid interfaces [30]. The major contributors to this

capacitive dispersion of such dispersion at solid interfaces involve adsorption of the anions, chemical inhomogeneity, roughness, and polycrystallinity level [30].

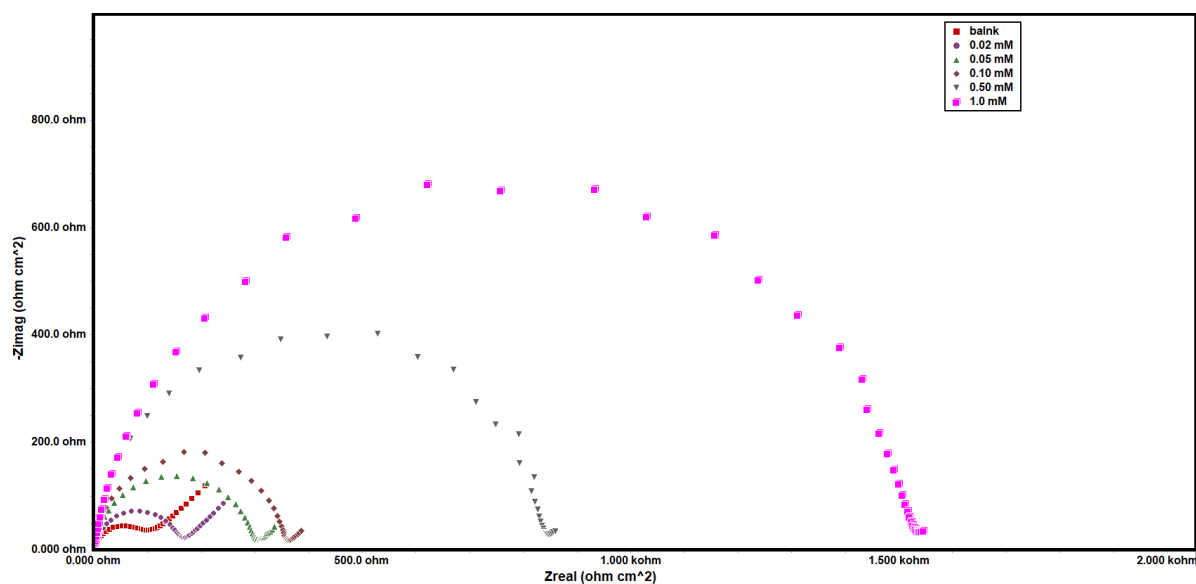


Figure.3a Electrochemical Impedance Spectra of copper corrosion in 1.0 M HNO_3 either without the presence or in the presence of Schiff base at various concentrations, when corrosion was observed.

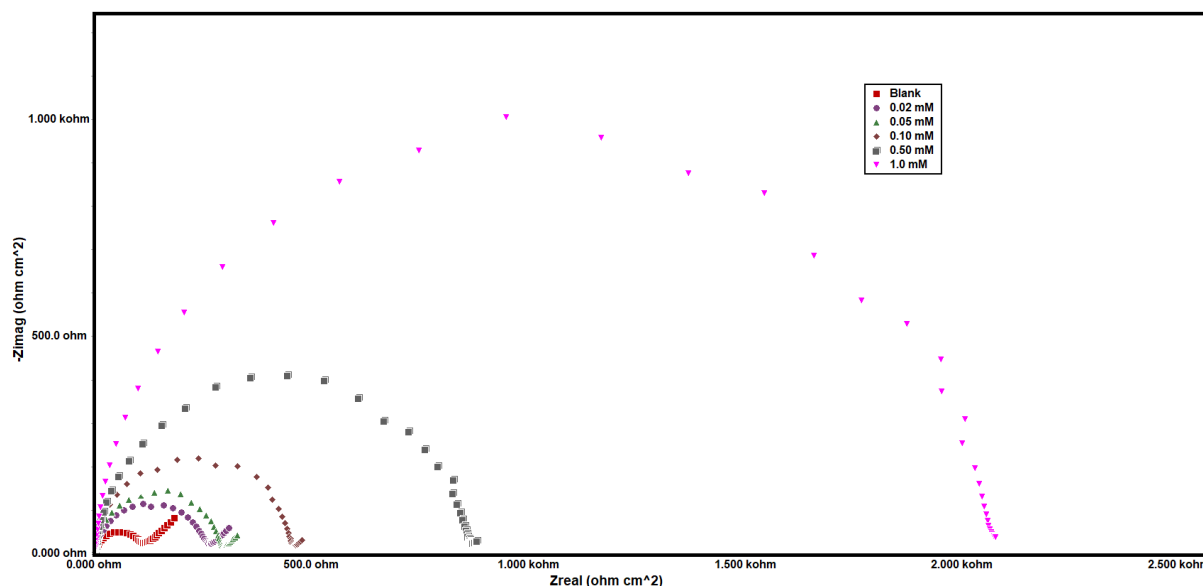


Figure.3b Electrochemical Impedance Spectra of corrosion of copper in 0.5M H_2SO_4 in the absence and presence of Schiff base at varying concentrations.

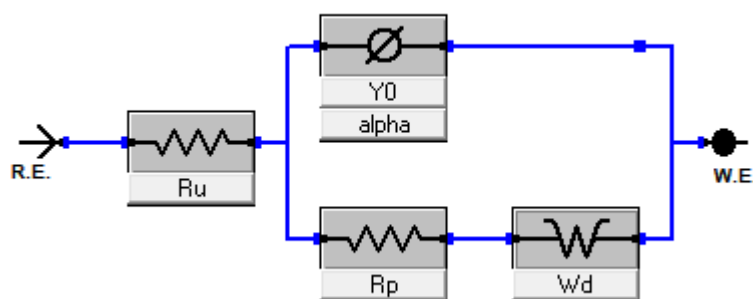


Figure.4 The circuit most likely to fit the experimental curves of impedance.

The analogous circuit shown in Fig. 4 [34,43], where R_p stands for the charge transfer resistance, W_d for the Warburg impedance, and R_u for the solution resistance, is used to examine the impedance

spectrum that shows a high frequency capacitive loop and the Warburg impedance at low frequencies. In aerated acid solutions, the diffusion of oxygen and subsequent reduction of that oxygen are thought to

be the causes of the formation of the Warburg impedance [44]. Since the capacitive loops are depressed semi-circles rather than standard semi-circles, which have the impedance function in the following form [45], a constant phase element (CPE) is used in place of the capacitive element to provide a more precise match [46].

$$Z_{\text{cpe}} = \frac{1}{Y_0(j\omega)^n}$$

(vi)

Y_0 = The admittance of the corrosive system at 1 rad/s, and n a constant ($-1 \leq n \leq 1$). In the case of $n=0$, CPE is a pure resistor, in the case $n=+1$ CPE is a pure capacitor, and in the case $n=-1$ CPE is an inductor [30].

The property of the CPE being nearly capacitive is observed as expounded in table 2 where the corroding electrode in the two acid media has the n value close to 1. The estimation of using the idealized model of crystallite alongside the capacitance (C_{id}) is determined by the CPE parameters [30, 34, 47].

$$C_{\text{id}} = \frac{Y_0 \omega^{n-1}}{\sin(\frac{n\pi}{2})} \quad (\text{vii})$$

where " ω " is the angular frequency at the imaginary portion of impedance's highest value. It is also evident that the formula [30,34,48] is used to compute the values of angular frequency and C_{dl} .

$$\omega_{\text{max}} = \left(\frac{1}{R_{\text{ct}} Y_0} \right)^{1/n} \quad (\text{viii})$$

$$C_{\text{dl}} = Y_0(\omega_{\text{max}})^{n-1} \quad (\text{ix})$$

The C_{dl} values reduced in both media but the R_p values rose with the increase in the concentrations of the inhibitors. When the inhibitor coverage of surface is increased, R_{ct} increases and hence, so does inhibitor efficiency as concentration increases. Replacement of water in the adsorption of the molecules of the inhibitor at the metal/solution interface led to a drop in the local dielectric constant, and/or an increase in the thickness of the electrical double layer, which explains the decline in the values of C_{dl} [30,34,49]. Corrosion was reduced to a small proportion as a result of slow substitution of the molecules of water at the metal surface with that of inhibitors [30,50].

3.3 Potentiodynamic Polarization Studies

It is evident from the potentiodynamic polarization curves shown in Figures 5a and 5b that the anodic and cathodic Tafel slopes change when an inhibitor is added to the corrosive medium, demonstrating the impact of Schiff base molecules on the corresponding processes. But in the cathodic curve, the influence was noticeably stronger. Since the cathodic polarization regions are nearly parallel to one another, activation control over the hydrogen development is evident [34,51]. The cathodic curves rightward shift indicates that cathodic reactions are primarily responsible for the acceleration of corrosion, in accordance with corrosion theories [34].

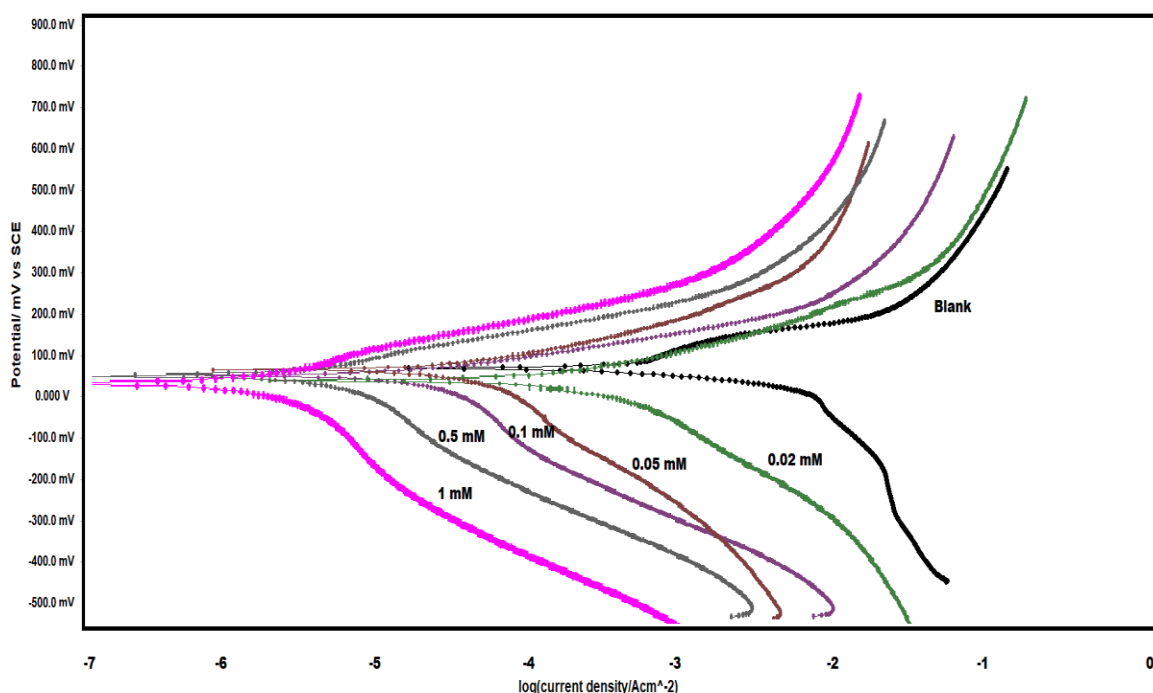


Figure.5a Tafel plots of copper corrosion in 1 M HNO₃ in presence and absence of Schiff base in various concentrations.

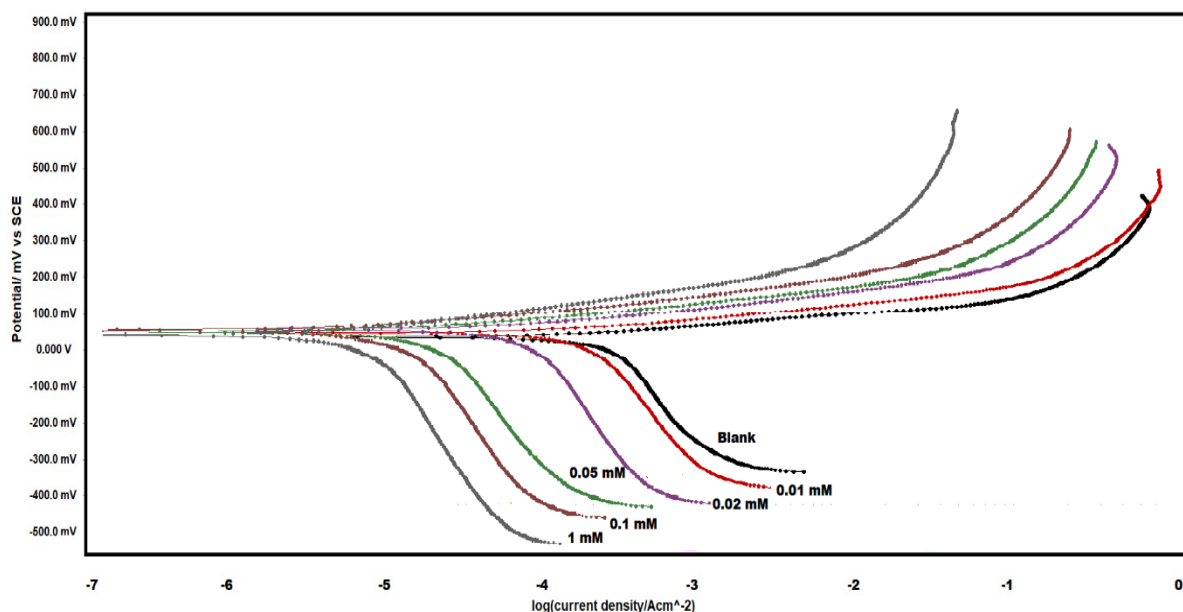
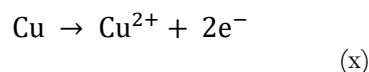


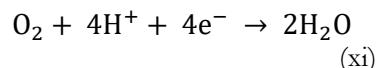
Figure.5b Tafel plots of copper corrosion in 0.5 M of H₂SO₄ in the absence and presence of Schiff base at varied concentration.

As a rule an inhibitor can be cathodic or anodic with the change of E_{corr} exceeding 85 mV/ E_{corr} of the uninhibited solution [30,34,52]. Further because the largest change of E_{corr} in the present research is lower than 85 mV when the inhibitor is added in both media, the inhibitor can be classified as mixed [34,51]. The fact that the slope at the anodic side of the Tafel polarization curves is not sharp may indicate that copper surface is not forming a passive layer. This is a fact observable in the Pourbaix diagram of copper as on acidic environment, the copper is reduced into Cu^{+2} , and it never forms its oxide film [51]. Dissolution of Cu into Cu^{+2} did not take place in deaerated acidic solutions but dissolves in dissolved oxygen giving rise to the Cu^{+2} ion by the following reaction:

Anodic reaction:



Cathodic reaction:



The diffusion process of the soluble Cu^{+2} ions, the location of which is at the outer plane of the Helmholtz plane into the bulk solution, controls the copper dissolution [51].

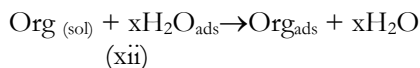
The electrochemical parameters that are seen in Table 3 include the corrosion potential (E_{corr}), the corrosion current (i_{corr}), anodic and cathodic Tafel slope, and inhibitor efficiencies. The β_c and β_a vary with rising inhibitor concentration and a stronger effect on slowing down the cathode reduction rate is observed. This implied that the Schiff base reduced the cathodic reaction compared to the anodic reaction.

Table.3 Corrosion of copper in 1.0 M HNO₃, and 0.5 M H₂SO₄ acid solutions in the absence and presence of Schiff base at various concentrations in electrochemical parameters.

Medium	Concentration (mM)	E_{corr} (mV)	I_{corr} ($\mu\text{A cm}^{-2}$)	β_a (mV)	β_c (mV)	IE%
HNO ₃	Blank	63.12	388.4	48.15	130.1	
	0.02	56.14	236.9	92.44	133.4	39
	0.05	59.07	147.6	137.2	139.8	62
	0.1	30.27	93.22	155.7	158.5	76
	0.5	35.88	46.61	181.3	166.4	88
	1	12.36	19.42	195.6	170.2	95
H ₂ SO ₄	Blank	15.3	69.42	104.4	57.2	-
	0.02	29.6	29.85	135.4	78.51	57
	0.05	63.3	26.38	139.8	94.32	62
	0.1	43.8	13.19	168.1	102.2	81
	0.5	52.6	6.942	172.5	128.6	90
	1	4.79	4.165	179.6	154.7	94

3.4 Adsorption Isotherm

Adsorption of the Schiff base molecules on the metal surface can be regarded as a replacement reaction where an organic compound (org_{sol}) from the aqueous phase replaces the water molecules bound to the metal surface ($\text{H}_2\text{O}_{\text{ads}}$) [30,34]



Where x is the number of molecules of water that is replaced by one 1,3-bis(pyridin-2-

methylmethylene)thiourea molecule adsorbed. The R_{ct} values were used to obtain the surface coverage of the inhibitor (θ) by using the equation,

$$\theta = \frac{R'_{\text{ct}} - R_{\text{ct}}}{R_{\text{ct}}} \quad (\text{xiii})$$

Where R_{ct} and R'_{ct} are the values of charge transfer resistance of the inhibited and uninhibited solution, respectively.

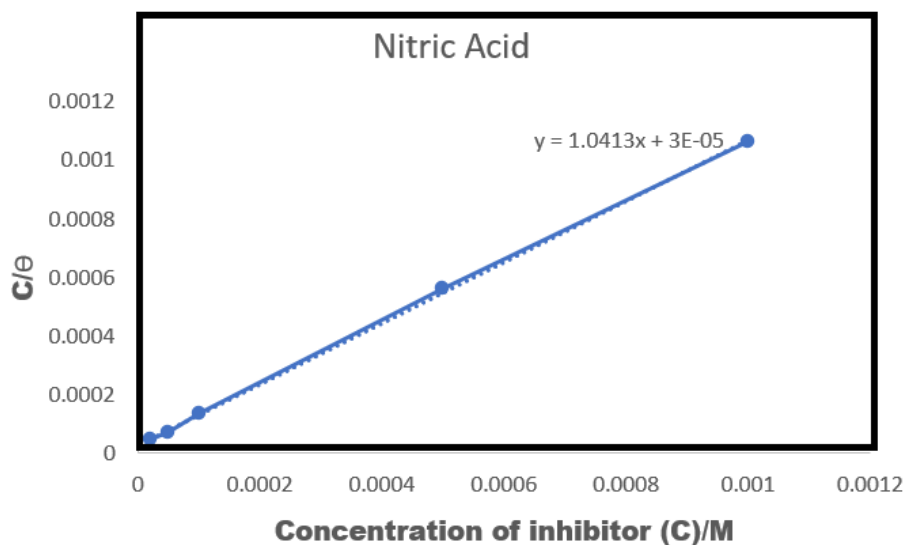


Figure.6a Isotherm of adsorption of 1, 3-bis (pyridin 2-ylmethylene) thiourea on copper in 1.0 M acid solution of HNO_3 .

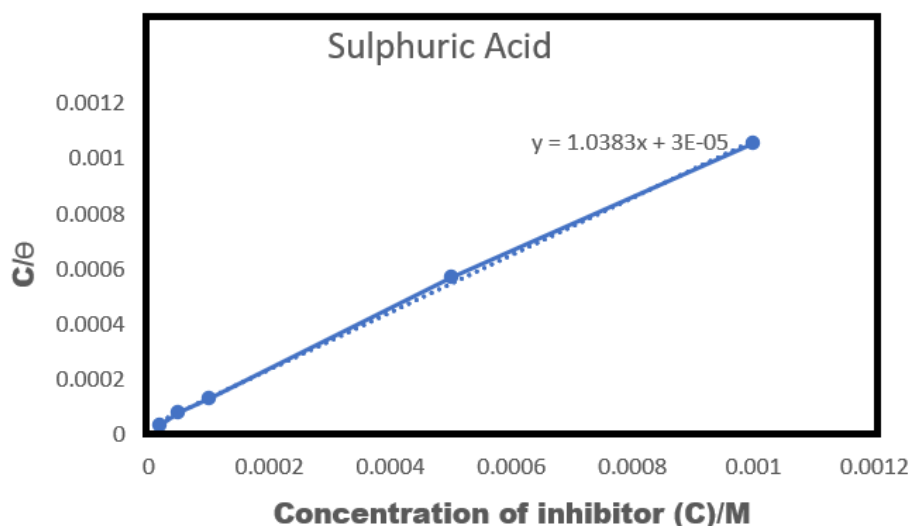


Figure.6b Adsorption isotherm and adsorption of 1,3-bis(pyridin-2-ylmethylene) thiourea on copper in 0.5 M H_2SO_4 acid solution.

Analysis indicated that the adsorption adhered to the Langmuir isotherm, represented by the equation [30,34,53],

$$C_{\text{inh}}/\theta = C_{\text{inh}} + 1/K_{\text{ads}} \quad (\text{xiv})$$

and the graphs are shown in Figures 6a and 6b. C is the concentration of the inhibitor and θ is the surface coverage and K_{ads} is the adsorption equilibrium constant. To calculate the standard free energy of adsorption, K_{ads} value is calculated by using the graph

of C vs C/θ at constant temperature in which the same is used and the following formula can be used [30]:

$$K_{\text{ads}} = 1/55.5 \exp^{-\Delta G_{\text{ads}}/RT} \quad (\text{xv})$$

Where, R is the universal gas constant, T is the thermodynamic temperature, 55.5 is concentration of water in the solution in mol L^{-1} . The obtained plots showed a linear nature at which the correlation coefficients were more than 0.9. The value of ΔG_{ads} in 0.5M sulfuric and 1.0M nitric acid solutions respectively are -36.09 and -36.10 kJ mol^{-1} at room temperature. The negative ΔG_{ads} values denote the spontaneous adsorption of 1, 3-bis(pyridin-2-ylmethylene)thiourea molecules to the metal surface.

In general, the $-\Delta G_{\text{ads}}$ less than 20 kJ mol^{-1} are considered physisorption and greater than 40 kJ mol^{-1} correspond to chemisorption. When the range of $-\Delta G_{\text{ads}}$ is between 20 kJ mol^{-1} and 40 kJ mol^{-1} it means that there is a physisorption and chemisorption [30,34,54].

3.5 Effect of Temperature

The Arrhenius graphs can be seen in Figures 7a and 7b. Examination of Table 4 indicates that the E_a values obtained in the presence of Schiff base are lower than those in the uninhibited solution. The reduction in E_a with the introduction of the inhibitor in both media indicates that the inhibitor molecules are chemisorbed onto the metal surface [55-57].

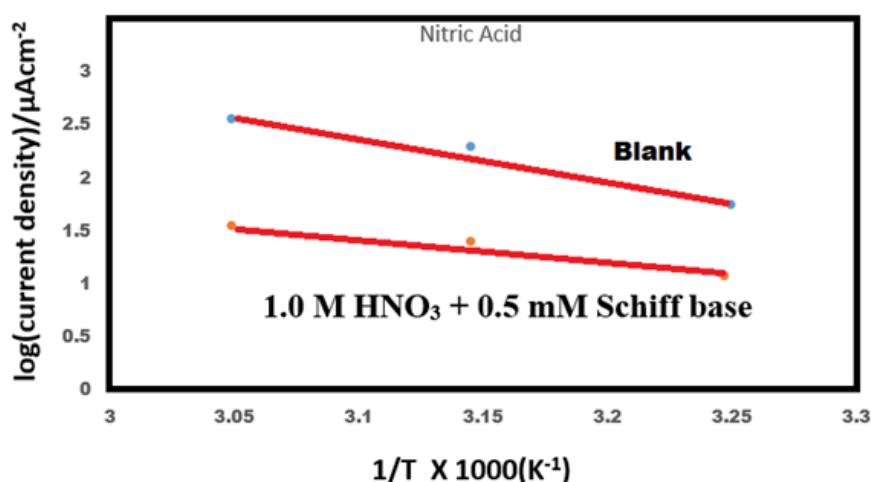


Figure.7a Arrhenius plots are of the corrosion of copper in the absence and presence of Schiff base in the copper in 1M HNO₃ acid solution.

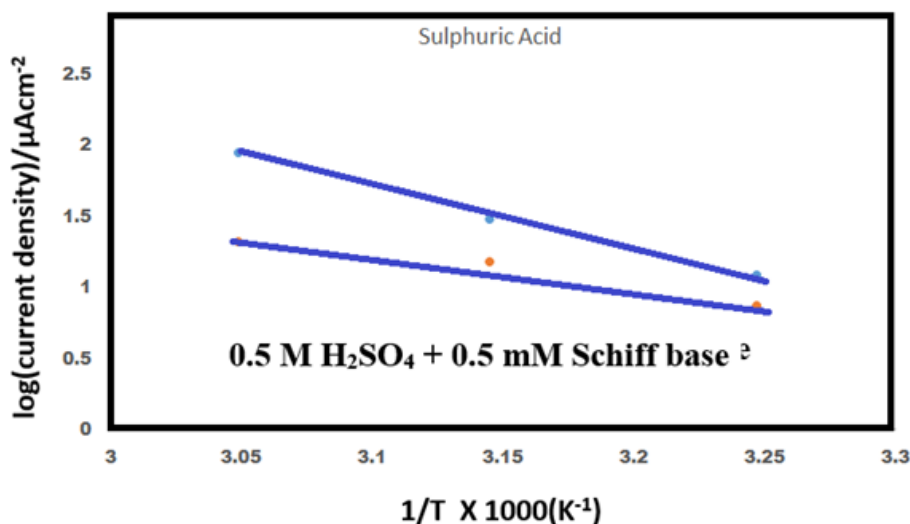


Figure.7b Arrhenius graphs corrosion of copper in 0.5M H₂SO₄ acid solution with and without Schiff base.

Table.4 Thermodynamic parameters of copper corrosion in 1.0M HNO₃ and 0.5M H₂SO₄ acid solutions in the absence and presence of Arrhenius plots explained the presence of Schiff base in the solution.

Medium	Inhibitor concentration	K _{ads} (M ⁻¹)	ΔG _{ads} (kJ/mole)	ΔH (kJ/mole)	ΔS (kJ/mole)	E _a (kJ/mole)	R
HNO ₃	Blank	-	-	-65.4	-24.5	66.21	0.981
	0.5mM Schiff Base	34610	-36.10	-44.1	-89.3	47.65	0.991
H ₂ SO ₄	Blank	-	-	-70.5	-40.2	78.16	0.990
	0.5mM Schiff Base	34710	-36.09	-25.4	-144.7	29.13	0.985

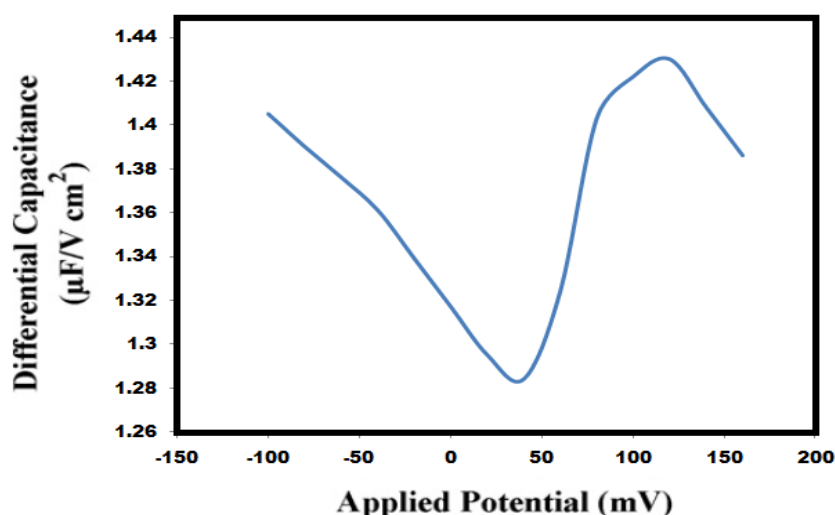
3.6 Potential of Zero Charge

The incorporation of the organic **compounds** on the metallic surface mostly relies on the surface charge and the charge or the dipole moment of the inhibitor molecule and other **charged species** specifically adsorbed on the metal surface [30,34]. Open circuit potential of the metal gives the surface charge of the metal [34]. Figures 7a, 7b, 7c and 7d depict the correlation of the applied DC potential and the double layer capacitance. The table 5 shows the E_{ocp} and PZC data. Copper surface charge at the open circuit potential is estimated by using formula: $E_r = E_{ocp} - E_{pzc}$ (xvi)

Where E_r is rational corrosion potential determined by Antropov [30]. The surface charge of copper at the OCP in 0.5M solution of sulphuric acid and in 1.0M solution of nitric acid, was observed to be positive in the inhibited and the uninhibited solution. This is proposed to occur by the initial adsorption of the sulphate and the nitrate **ions** upon the surface of the metals, [36, 42,43] by the subsequent adsorption of protonated inhibitor-molecules on to the anions and, thus, inhibiting the dissolution of the metals, [34] and by the positive charge of the metal surface towards PZC in the acidic solutions.

Table.5 Overcharge on the copper metal surface of 1.0M HNO₃ and 0.5M H₂SO₄ acid solutions in the presence and absence of Schiff base.

Medium	E _{ocp} (mV/SCE)	PZC (mV/SCE)	Excess charge E _{ocp} - E _{PZC} (mV)
1.0M HNO ₃	+44.50	+41.5	+3.0
1.0MHNO ₃ + 0.5mM of Inhibitor	+51.20	+50	+101.2
0.5M H ₂ SO ₄	+2.60	-30	+32.60
0.5MH ₂ SO ₄ + 0.5mM of Inhibitor	+32.60	+30	+2.60

**Figure.7a** basis source Plot of differential capacitance versus Applied potential in YINO 3 acid solution, 1.0 M.

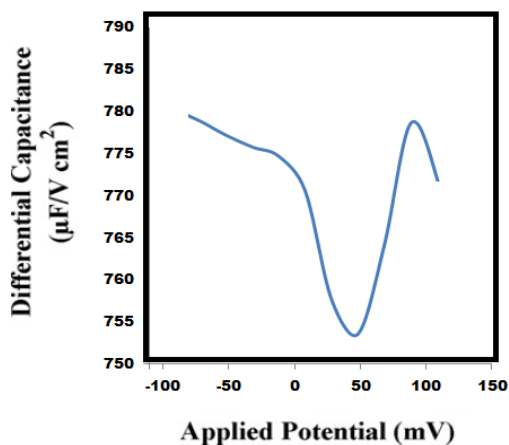


Figure.7b Differentiated capacitance versus Applied potential of copper in 1.0M HNO₃ acid solution in the presence of Schiff base.

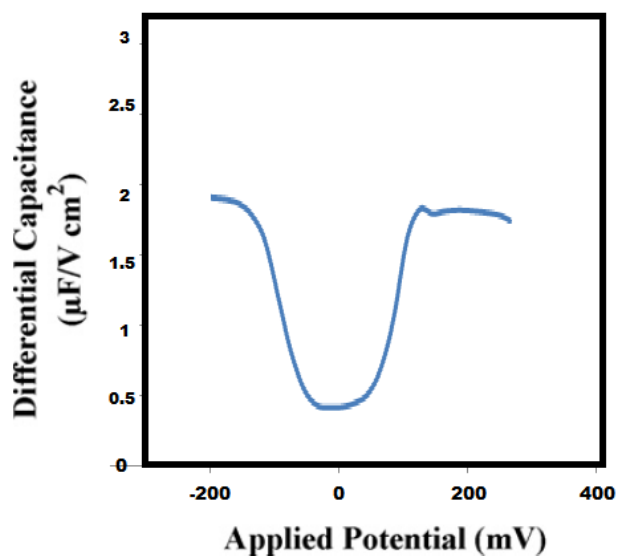


Figure.7c Differentiated capacitance vs Applied potential copper in 0.5M H₂SO₄.

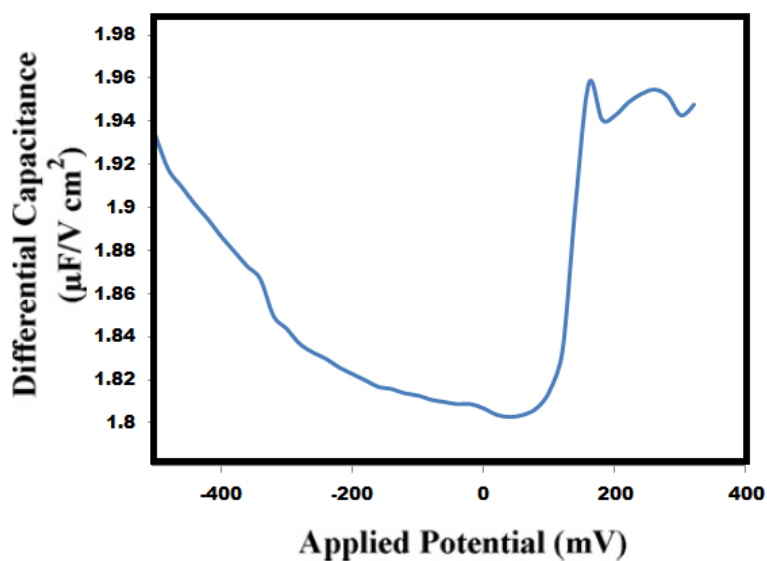


Figure.7c Differentiated capacitance vs Applied potential in copper solution in 0.5M H₂SO₄ acid in the presence of Schiff base.

3.7 Synergism Parameter

The impact of establishing two or more corrosion inhibitors within the corrosive environment may enhance the intervention of the inhibitors accelerating the effects of the corrosion inhibitors rather than just cumulative. In general, addition of halide ions or other to a solution that has been inhibited exhibits synergistic effect when SI is greater one and antagonistic when SI is not **greater than one**. The halide ions are believed to enhance the adsorption of organic cation by offering intermediary bonds to the positively charged metal surface and the positively charged end of organic inhibitor [58-60]. The synergist corrosion inhibition occurs because inhibition enhances its surface concentration through

three ways of enhancing ion-pair defect in both the organic cations and the halide anions. Table 6 presents the synergy parameters SI derived as a result of polarization experiment that was conducted on potassium chloride, potassium bromide as well as potassium iodide. The synergistic effect exists in the case in question and is represented by the presence of iodide ions and the antagonistic effect is represented by chloride and bromide ions, in both of the acidic environments. This increased effect of the iodide ion has been frequently explained by its big ionic radius, high degree of hydrophobicity and low electronegativity in contrast to the other halide ions that allow the attaching of the molecules of an inhibitor to the metal surface [40].

Table.6 The effect of halides ion on copper corrosion of 1.0 M HNO₃ and 0.5M H₂SO₄ acid solution in the absence and presence of Schiff base measured using Electrochemical impedance spectroscopy.

Medium	Inhibitor Concentration (mM)	R1	W	Y _O (X10 ⁻⁶) Ω ⁻¹ cm ⁻²	n	R _{ct} Ωcm ²	θ	S _θ
1.0M HNO ₃	Blank	1.124	24X103	871	0.92	91.1	-	-
	0.5mM NF	1.148	96X103	79.51	0.959	829.5	0.89	-
	0.5mM KCl	1.047	19X103	187.2	0.911	329	0.72	-
	0.5mM NF + 0.5mM KCl	2.113	49X103	168.4	0.918	135	0.33	-
	0.5mM KBr	1.006	15X103	266.5	0.912	236	0.61	-
	0.5mM NF + 0.5mM KBr	1.993	69X103	199.7	0.945	335	0.73	-
	0.5mM KI	1.021	12X103	101.3	0.958	129	0.29	-
	0.5mM NF + 0.5mM KI	2.204	83X103	86.4	0.934	1236	0.93	2.5
	0.5mM KI	1.021	12X103	101.3	0.958	129	0.29	-
0.5M H ₂ SO ₄	Blank	1.357	35X103	426.1	0.953	102.7	-	-
	0.5mM NF	1.441	121X103	69.23	0.955	1029	0.90	-
	0.5mM KCl	1.192	22X103	349.1		197.2	0.48	-
	0.5mM NF + 0.5mM KCl	2.126	122X103	133.5	0.954	336	0.69	-
	0.5mM KBr	1.029	21X103	263.4	0.943	160.2	0.36	-
	0.5mM NF + 0.5mM KBr	1.661	147X103	262.1	0.981	532	0.81	-
	0.5mM KI	1.026	19X103	158.4	0.966	179.3	0.43	-
	0.5mM NF + 0.5mM KI	1.228	152X103	132.4	0.954	2125	0.95	6.6
	0.5mM KI	1.026	19X103	158.4	0.966	179.3	0.43	-

3.8 Quantum Chemical Calculations

The optimized structure of 1, 3-bis (pyridin-2-ylmethylene)thiourea is provided in Figure 8. One of the approaches in which the data concerning the geometries and distributions of electrons of a certain molecule was calculated in the correct way was the density functional theory(DFT). Table 7 presents calculated parameters of theoretics that offer the data concerning the reactivity of the inhibitor. Each high occupied molecular orbital (E_{HOMO})

Molecular orbital) represents the fact that an electron can be given off by a molecule. Hence, an increase of E_{HOMO} value means that a tendency to give out electrons increases, contributing to the increase in the efficiency of the adsorption of the inhibitor onto the metal surface [34,62]. E_{LUMO} on the other hand is the capacity of the molecule to take electrons. Under this, the affinity of the metal surface with the inhibitor increases with increasing HOMO and a decreasing LUMO energy.

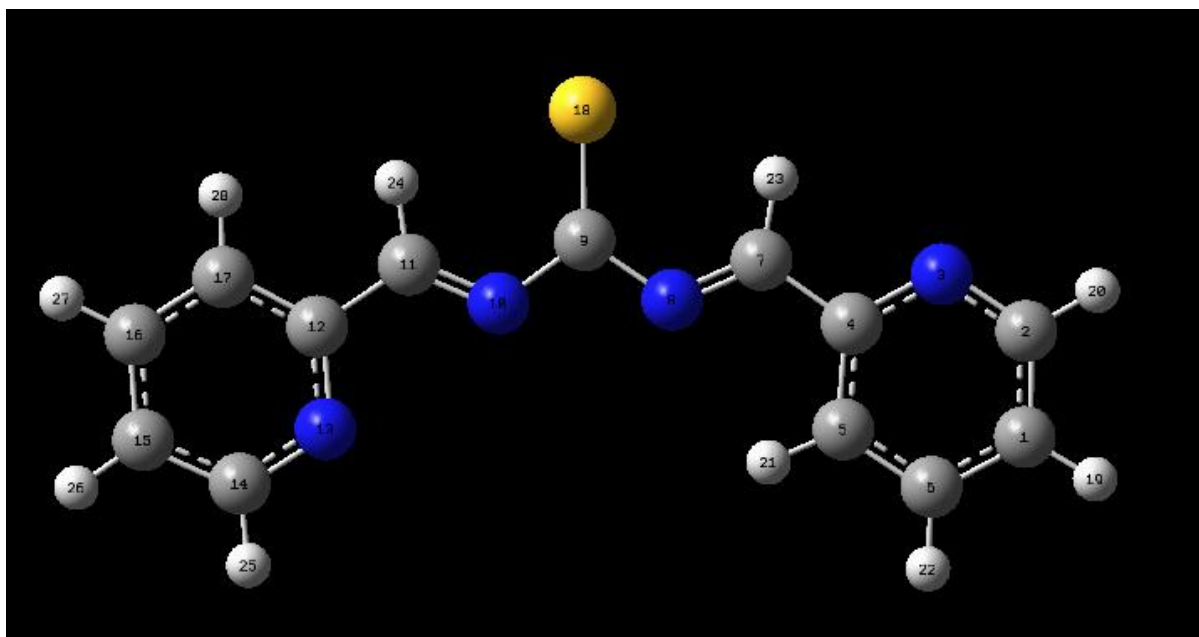


Figure.8 Streamlined structure of the produced synthesized Schiff base 1,3-bis((pyridin-2-ylmethylene)thiourea.

Table.7 Quantum chemical Parameters obtained in the 1,3-bis((pyridin-2-ylmethylene)thiourea and the protonated ions.

S. No	QC Parameters	Unprotonated Schiff base	Protonated at N-10 or N-8	Protonated at ring nitrogen (N-3 or N-13)	Protonated at N-8 and N-10
1	Total energy (eV)	-1980.01038	-1993.7371	-11370.31216	-2007.15113
2	E_{HOMO} (eV)	-9.254	-7.708	-7.236	-7.102
3	E_{LUMO} (eV)	-1.081	-0.046	-0.139	-0.05
4	ΔE (eV)	8.173	7.662	7.097	7.052
5	Ionization potential (I/eV)	9.254	7.708	7.236	7.102
6	Electron affinity (A/eV)	1.081	0.046	0.335	0.05
7	Hardness (η /eV)	4.086	3.831	3.526	3.450
8	Softness (S/eV)	0.122354093	0.130514226	0.144906535	0.141803744
9	Electronegativity (χ)	5.1675	3.877	3.7855	3.576
10	Electrophilicity (ω)	3.267228221	1.961776168	2.076512136	1.813354509
11	Fraction of electrons transferred (ΔN)	0.224213875	0.407595928	0.465802058	0.485536018

The activation energy (ΔE) between HOMO and LUMO of the molecule is one of the important aspects, which can determine the capacity of the inhibitor molecules to adsorb onto the metallic surface. ΔE decreasing corresponds to an increase in the reactivity of the molecule, and this leads to the increased efficiency of the inhibition of molecule[34]. A review of Table 7 depicts that the E_{HOMO} value is large when the Schiff base is protonated at N-8 and N-10 and E_{LUMO} value is the lowest at the neutral form. Consequently, the electronically charged state, which is the double protonated, is the one that is very

vulnerable to giving electrons to the atoms located at the surface and the neutral one is the one that is very vulnerable to receive electrons to the occupied orbitals of the atoms on the surface.

The electronic charge distribution of HOMO and LUMO of double protonated form of Schiff base is shown in Figure.9a and 9b. In the present work, the ΔE value is the lowest for the double protonated form and this indicated the greatest tendency of this form to get adsorbed onto the copper metal surface.

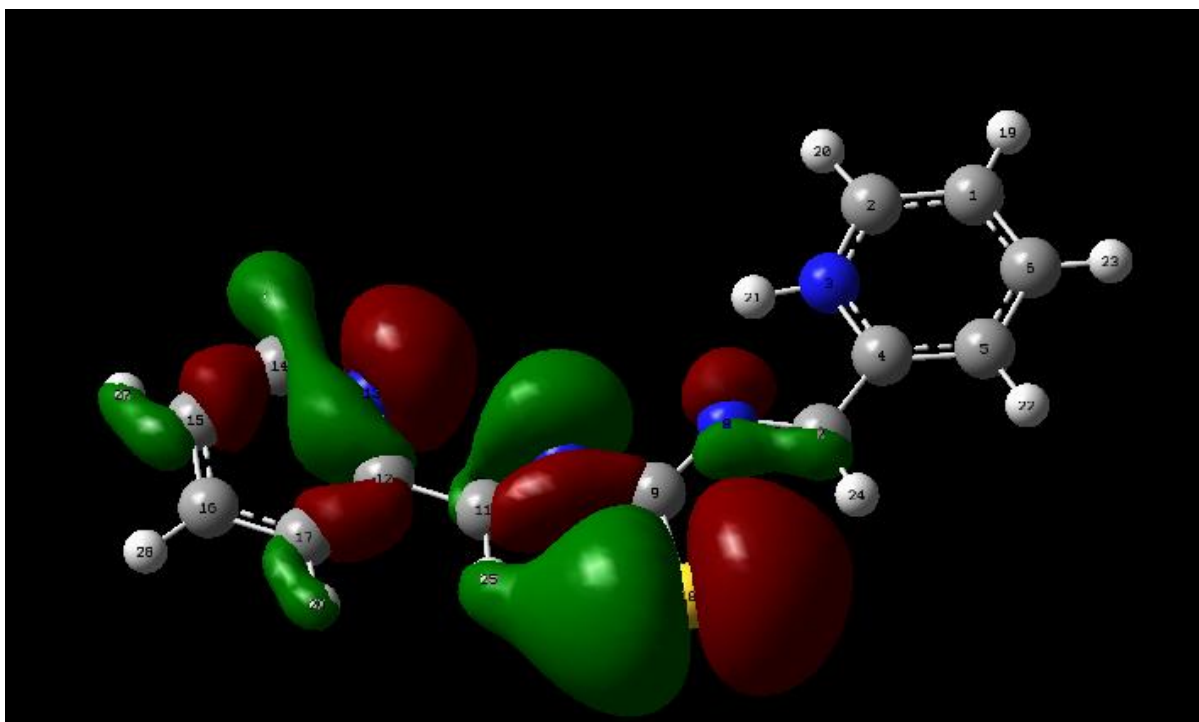


Figure.9a Electronic charge distribution in HOMO of double protonated (N-8 and N-10) form of 1,3-bis((pyridin-2-ylmethylene)thiourea.

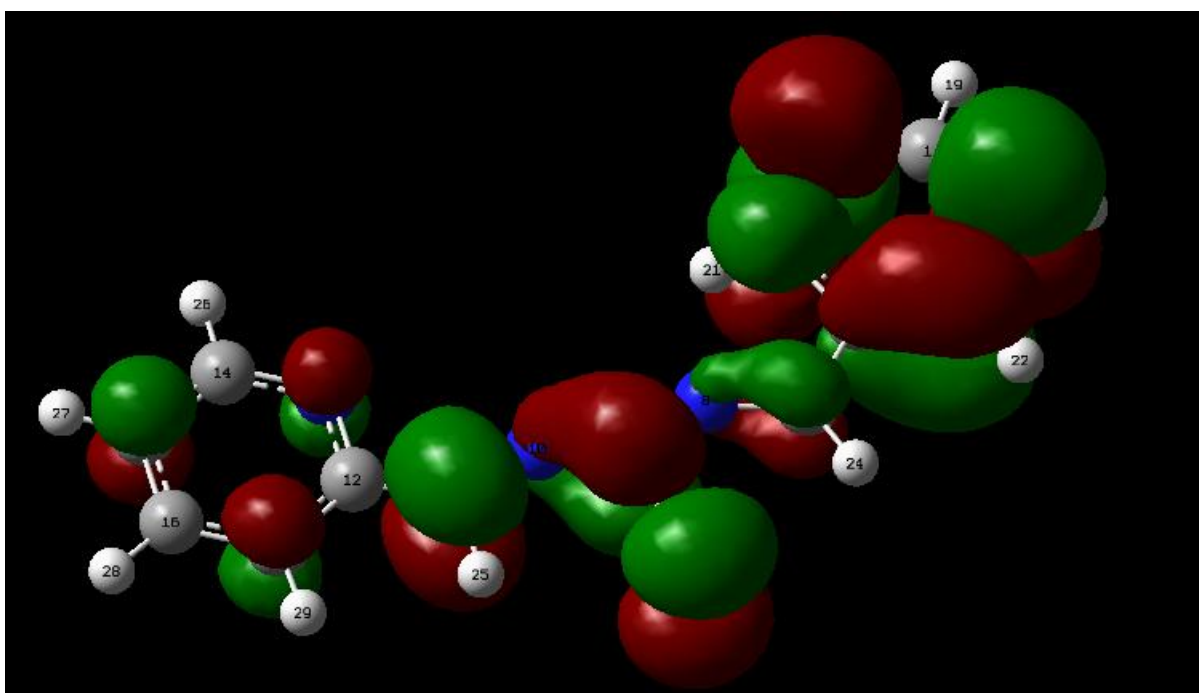


Figure.9b Electronic charge distribution in LUMO of double protonated (N-8 and N-10) form of 1,3-bis((pyridin-2-ylmethylene)thiourea.

An important indicator that is based on ionization energy of atoms and molecules is the chemical reactivity of atoms and molecules. The high ionization energy is associated with a high level of stability and inertia and the low ionization energy with a high level of reactivity of molecules and atoms [34]. The most protonated form is the double protonated form with the lowest low ionization energy that might be explained by greater efficacy.

A theoretical χ value of 4.48 eV/mol using the Pearson electronegativity scale and η of 0eV/mol using copper were used to calculate ΔN and gave it as shown in Table 7. The double protonated form has the highest value for fraction of electron transferred, which indicated that this doubly protonated form might have major contribution for the observed corrosion inhibition efficiency.

Hardness and softness are valuable values which are used to establish the stability and reactivity of

molecules. The energy gap in a hard molecule is a large energy gap whilst a soft molecule has a small energy gap [62]. The binding may be done at the part of the molecule with the highest value of softness (S) [62]. The inhibitor with the lowest global hardness among them is expected to have the most inhibition efficiency [62] meaning that the inhibitor shows the highest inhibition efficiency of the double protonated form compared to other forms.

4. Conclusion

The current study reached the following conclusions: (i) Within an environment of 1.0M nitric acid and 0.5M sulphuric acid, there is an increase in the efficiency of the inhibitor with the increase in the concentration of 1, 3bis(pyridin-2-ylmethylene)thiourea with an optimum inhibitor efficiency observed at an interval of 1.0mM. Findings derived show that the copper corrosion can be inhibited in both acidic conditions using 1,3-bis(pyridin-2-ylmethylene)thiourea.

(ii) It is also the same pattern in the calculated inhibition efficiencies of studies of electrochemical impedance, and potentiodynamic polarization.

(iii) 1,3-bis(pyridin-2-ylmethylene)thiourea exhibits almost **same** corrosion inhibition efficiency in both the acid media.

(iv) The potentiodynamic polarization curves showed that this molecule is a mixed type of inhibitor that is predominantly cathodic in nitric acid.

(v) Since 1,3-bis(pyridin-2-ylmethylene)thiourea is adsorbed on a metal surface, it is adsorbed according to Langmuir adsorption isotherm. The adsorption process is spontaneous as it has negative values of G_{ads} .

(vi) The free energy of adsorption is approximated and equals -40 kJ mol^{-1} ; this implies that the adsorption is performed using the electrical coulombic force of attraction and chemisorption. The fact that the value of the E_a is decreased when one has the inhibitor indicates the inhibitor is chemisorbed onto the metal surface.

(vii) PZC formation means the formation of sulfate and nitrate linking chains during the process of inhibitor adsorption to the metal surface at acidic conditions.

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